



MA 604: Introduction To Game Theory

Course Code	MA 604
Course Name	Introduction To Game Theory
Approval	5th Senate (Oct 2013)
Recoded from	MA 705 by the 30th BoA (Jun 2019)

Course Outline

Introduction to game theory, routing games and mechanism design; Strategies, costs, and payoffs; Prisoner's dilemma, Nash Equilibrium, Strategic games; Best response; Dominant strategies; Pure strategy v/s Mixed strategy Repeated games; Bayesian games Routing games; Selfish routing; Quantifying inefficiency of equilibria; Price of Anarchy; Social optimum; Price of stability; Scheduling games Population games; Evolutionary game theory; Evolutionary stable strategy; Replicator dynamics Non-cooperative games; Cooperative game theory; Nash bargaining Mechanism design, Algorithmic mechanism design, Distributed algorithmic mechanism design